Ethan Schledewitz-Edwards 100908840
Andrew Hoult 100866035

# Project Timeline Overview

## Timeline

Figures 1 and 2 showcase this project's full timeline. Progress on the project started **after** the third week of lectures.

**Figure 1**
*The first five weeks of the project's timeline*

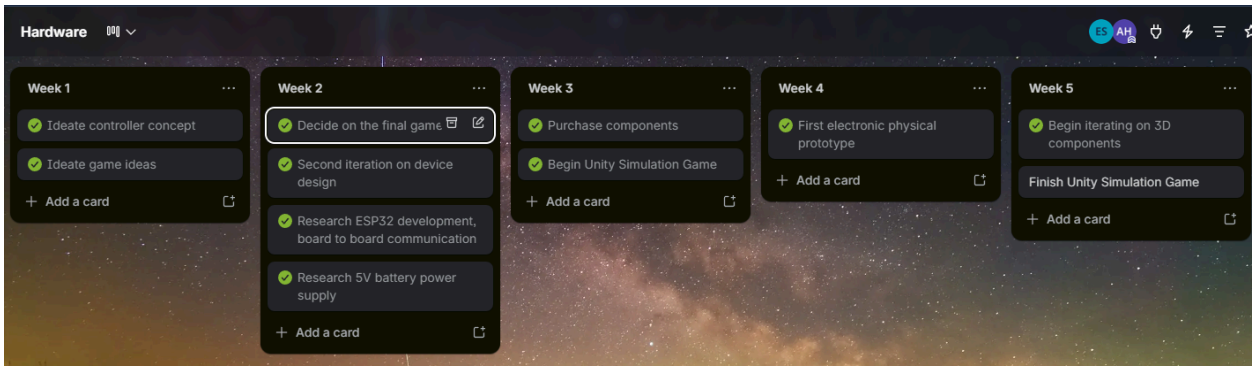
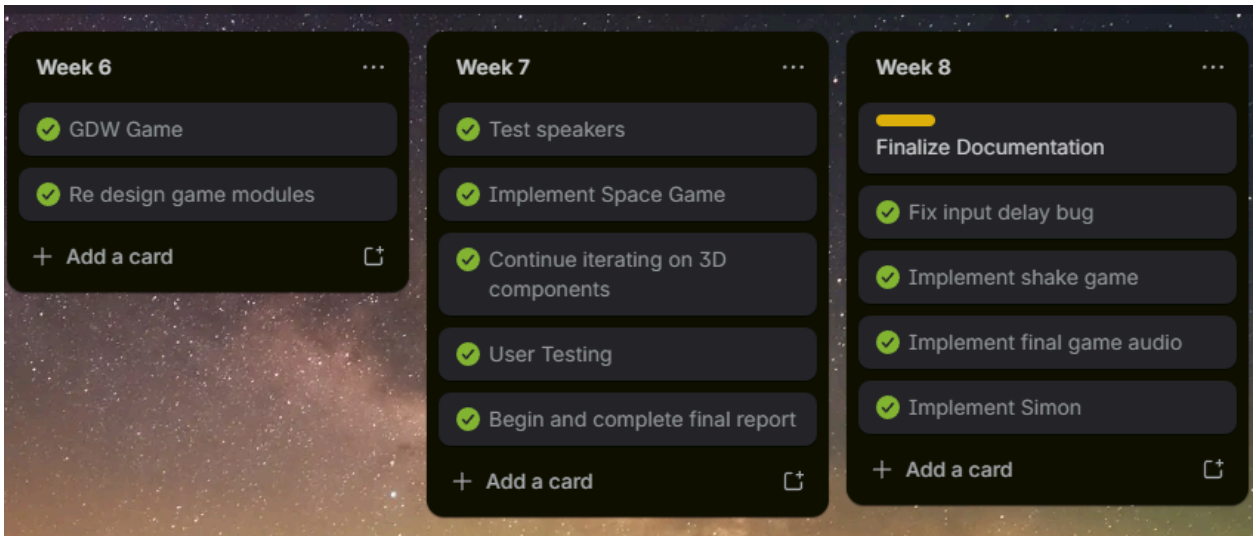



**Figure 2**
*The remaining three weeks of the project's timeline*




### Week 1

Week one began with design thinking activities, conceptualizing multiple controllers we would like to create. In-class activities allowed us to gather feedback on our ideas from other groups.

**Week 2**

Based on our findings in week one, the final concept for our controller was selected and iterated on. The group we consulted the week prior recommended that we include unconventional input methods into our controller to create an element of surprise. Based on the finalized concept, research was conducted to determine which electronic components and materials we needed to purchase to build the controller.

**Week 3**

Based on the research from the previous week, electronic components were purchased from Amazon. Additionally, a Unity-based simulation of the project began development. The simulation would have allowed us to test the flow of game modules and how they would behave.

**Week 4**

Unfortunately, due to a heavy course load, the Unity prototype was canceled as we deemed it unfeasible. However, the parts ordered the previous week were arriving. Using these components, the initial physical electronic prototyping began using a breadboard.

**Week 5**

As previously mentioned, the digital Unity prototype had been canceled in week 4. However, in week 5, 3D components were beginning to be iterated on using Fusion 360. These designs focused on housing for the individual components. This week included housing for the Keypad, LCD, and breadboard.

**Week 6**

Week six mostly focused on improving our GDW game before Milestone 3. However, programming for the controller prototype began, LCD issues were rectified, and an LCD interface was created.

**Week 7**

The LCD issues are somewhat resolved, the first game is complete, all components are integrated and functioning, and an API for each component has been created. User testing is conducted in the game lab. The final report is written and completed using the data gathered from the user testing.

**Week 8**

Minor bugs were fixed, and all game modules and code were completed before Tuesday (LevelUp Convention). Finally, the project was submitted on Wednesday morning.

## Work Breakdown

Andrew: Assembled electronics, 3D printed parts and iterated design, programmed microcontroller games.

Ethan: I worked on the concept, design, canceled Unity Prototype, some 3D component designs, and user testing of the project along with Andrew. I also handled all of the technical documentation.

Overall, we feel that the workload was evenly distributed between us, 50% each.